Leaf Disease Detection Using Deep Learning and using IoT

Abstract

This project presents a hybrid IoT and Deep Learning framework for precise, automated plant disease detection. IoT sensors gather real-time environmental data, while CNNs classify plant diseases accurately. The system is energy-efficient, scalable, and reduces manual effort—boosting productivity and promoting sustainable, cost-effective farming.

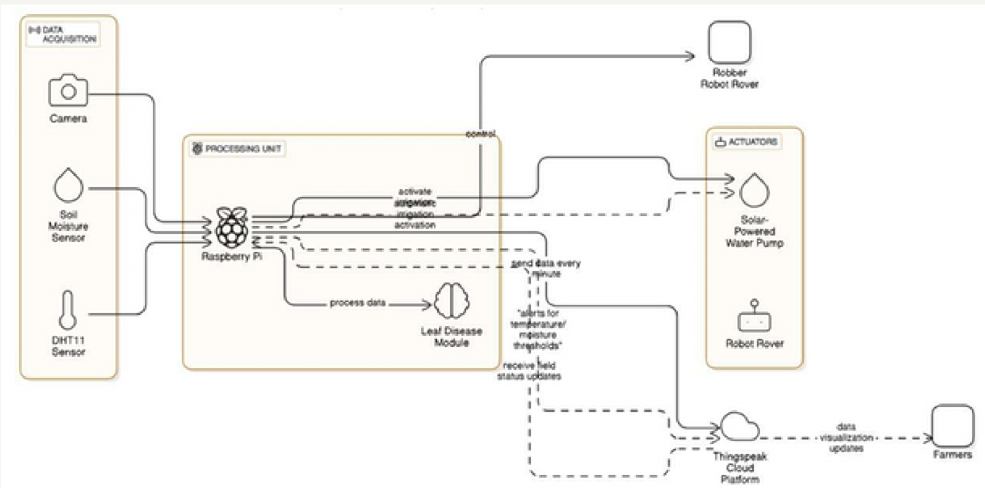
Software Requirements

- **OS**: Windows 10/11 or Ubuntu 20.04+ (Model Training), Raspberry Pi OS (IoT Deployment)
- **Language**: Python 3.7+
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV
- **Frameworks**: TensorFlow/Keras (CNN), Scikit-learn (ML), Flask
- **Cloud & DB**: ThingSpeak (IoT Data Storage), SQLite (User Login)

Hardware Requirements

- **Processor**: Intel i5/i7 (8th+ Gen) / Ryzen 5/7, 8–16GB RAM, 256–512GB SSD
- **IoT Unit**: Raspberry Pi 4 (4GB), 32GB+ MicroSD, 5V/3A Adapter
- **Sensors**: DHT11/DHT22, YL-69, NPK, Rain Sensor
- **Camera**: Pi Camera Module v2
- **Actuators**: 5V Relay, Servo Motor
- **Connectivity**: Wi-Fi / Ethernet (ThingSpeak Upload)

Architecture



Results